Design and kinematic analysis of A rigid-elastic hybrid gripper with the capability of force sensing

Hao Yuan[1,3], Siyue Yao[1,3], Lingyu Kong[4], Hao Wang[1,2,4], and Genliang Chen[1,2,3,*]

[1] Shanghai Key Laboratory of Digital Manufacture for Thin-walled Structures, Shanghai Jiao Tong University, Shanghai 200240, China
[2] State Key Laboratory of Mechanical Systems and Vibration, Shanghai Jiao Tong University, Shanghai 200240, China
[3] Shanghai Key Laboratory of Intelligent Robotics and Meta Robotics Institute, Shanghai Jiao Tong University, Shanghai 200240, China
[4] Institute of Marine Equipment, Shanghai Jiao Tong University, Shanghai 200240, China

*Corresponding author's e-mail: leungchan@sjtu.edu.cn

Abstract: Gripping force sensing is of vital importance in the application of industrial grippers. However, the existing gripping force sensing methods of grippers mainly depend on the current monitor of the actuating motor, which lacks stability and compliance. This study presents a novel gripper with elastic beams embedded in its actuating mechanism systems. Using the proposed discretized modeling method and local deformation sensing, the maximum force sensing error is 2.5 N with the maximum gripping force over 45 N. This design improves the compactness and stability of the force-sensing system. The gripper shows a profound usage in the industrial automation field.

1. Introduction

To acquire the gripping force, several methods have been proposed, including current sensing, SEA actuator angle sensing [1-2], pressure sensing [3], resonant sensing [4], SMA [5], vision sensing [6-10] and strain sensing [11-17]. Current sensing is widely used in the industry for sensing the grabbing force. However, current may be disturbed by several factors like motor vibration. The angle of the actuator is widely used in SEA for sensing force and displacement. A fin-shaped flexible gripper with SEA has been proposed with the capability of sensing grabbing force by angle sensors [1]. A humanoid gripper with SEA named PG2 has been proposed to implement force control over the grasped objects by monitoring the encoder [2]. Pneumatic grippers have arisen great attention for adaptivity in grasping tasks. A bellow-type soft actuator was proposed and its tip force could be obtained through a pressure monitor, which was put into use in a force-sensing gripper [3]. SMA combined sensing and actuation functions with a special alloy. A gripper made up of SMA was proposed with force sensing ability [5]. Vision-force sensing is a new type of force-sensing method. Fin-shaped flexible gripper was equipped with marking points, which provide displacement information for contacting force analysis [6]. Deformation was obtained by camera for force analysis in compliant grippers [7-9]. Optical fiber was also used for force and contact sensing [10], which is also considered a vision methodology for force sensing. A strain gauge is widely used for measuring deformation for high precision and low hysteresis, and the strain information could be put into physical models to calculate tip force. Strain gauges were used in a hydraulic soft gripper for underwater application while sensing the gripping force [11]. Tactile sensors were deployed on the contact region of a soft gripper to monitor the contact force and location [12-14]. Gripping force could also be sensed by physical analysis methods with strain gauges away from the contact region [15-17].

Among all the gripping force sensing methods, strain gauges are known for their compactness, stability, and accuracy. To make full use of it, a novel gripper with a rigid-elastic hybrid structure is

Content from this work may be used under the terms of the Creative Commons Attribution 4.0 licence. Any further distribution of this work must maintain attribution to the author(s) and the title of the work, journal citation and DOI.
Published under licence by IOP Publishing Ltd

proposed in this research. Elastic beams equipped with strain gauges are put into use for transmitting the driving force from the actuator and sensing the gripping force by monitoring deformation. A kinematic model is proposed for analyzing the gripping length and gripping force. The gripper is mainly composed of a strain collection and control system, a motor driver system, and a mechanical system. This novel design improves the compactness of the flexible gripper, providing ideology for future design.

2. Mechanical Design and Architecture Characteristics

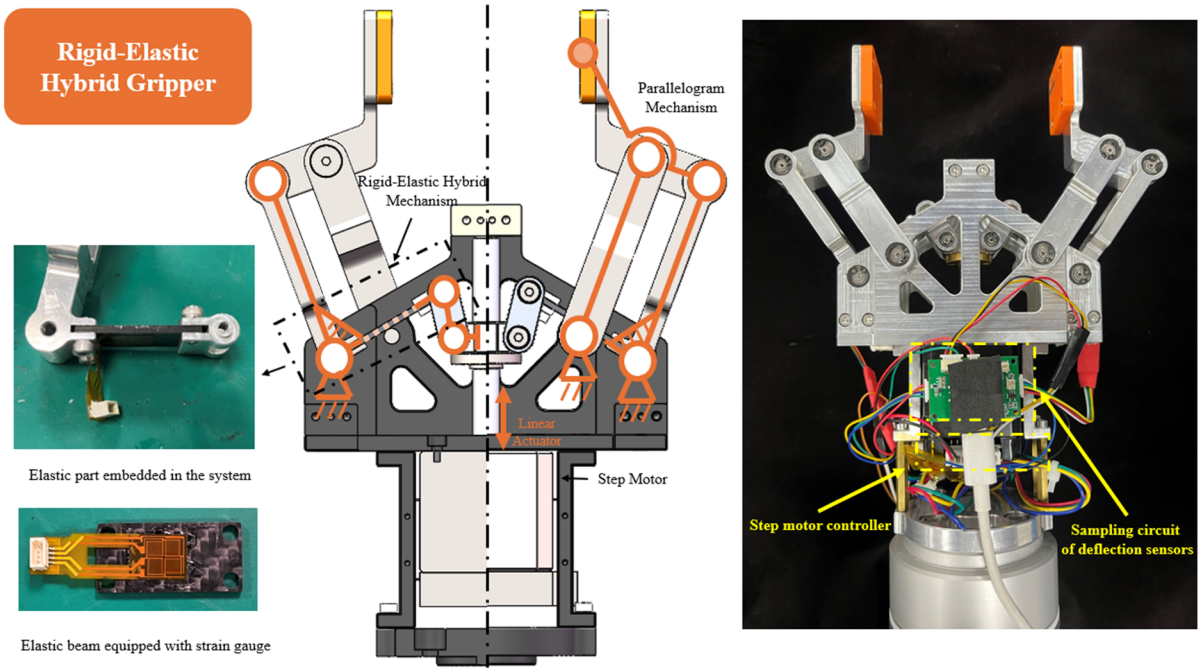


Figure 1. The mechanical design and electronic system of the gripper.

The mechanical design of the proposed gripper is shown in Figure 1. The rigid parallelogram is actuated by a rigid-elastic hybrid mechanical system. A 2-mm screw lead T-shaped lead screw is used to give linear actuation to the mechanical system, which is powered by a step motor. Thus, the input linear motion of the system could be obtained by monitoring the step of the motor. The elastic part is made of a carbon fiber plate, which is embedded in the rigid system as shown in Figure 1. The fiber plate is equipped with a strain gauge, enabling the gripper with the ability to sense local deformation, and therefore, the gripping force and fingertip position can be sensed. The gripper's tip is combined rigidly with one beam of the parallelogram mechanism.

The sampling circuit collects the data from the strain gauge in real time and transmits information through a USB cable. The maximum sampling rate of the deformation monitoring system is 3, 600 Hz. The motor is powered and controlled by a step motor controller, which will precisely control the rotation angle of the motor with feedback control. Thus, the motor angle used in the later model is considered to be the true value of the rotating angle of the lead screw. The electronic system endows the gripper with the ability to sense and control.

3. Elastostatics Modeling of Hybrid System and Stiffness Calibration

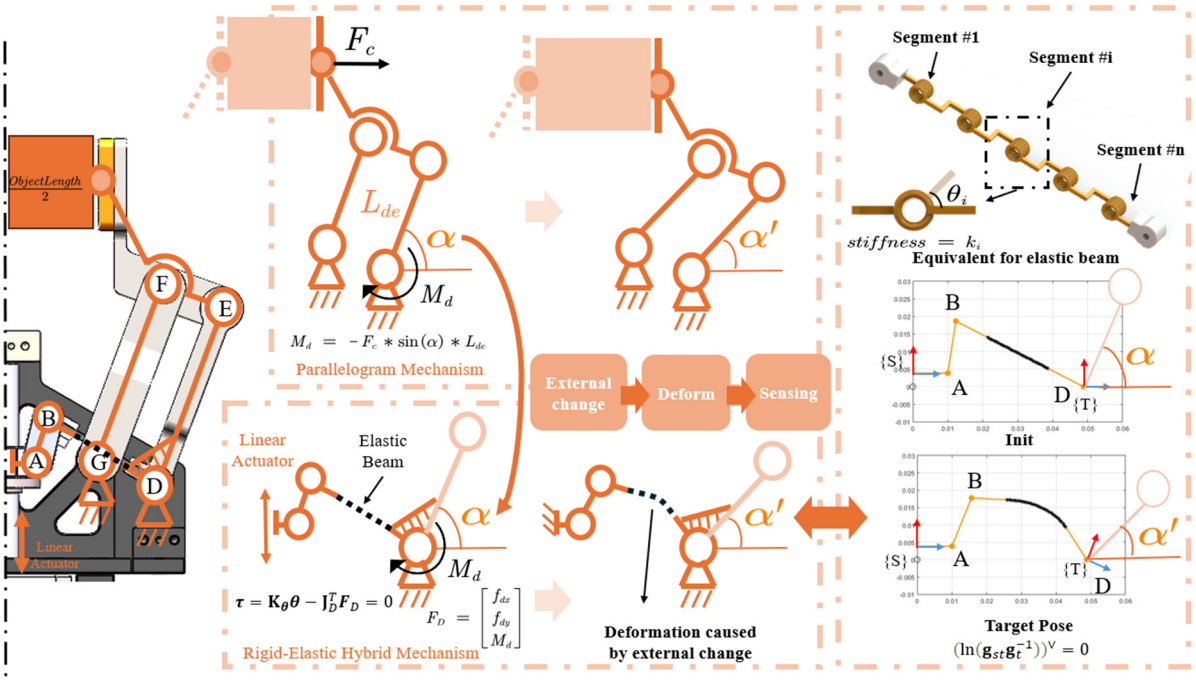


Figure 2. Elastostatic modeling of the gripper's mechanism.

As is shown in Figure 2, the rigid-elastic hybrid mechanism is separated into two mechanical systems. One of them is the parallelogram mechanism, whose position relation between the acuate angle α and the tip position could easily be solved. The model only considers the gripping force, so that the relation between the acuate torque and the gripping force could also be solved.

For the rigid-elastic hybrid mechanism, a discretization-based model is introduced. The elastic beam is segmented into elastic revolute joints with the same value of stiffness so as to depict the elastostatic characteristics of the mechanical system and be put into the robotic analysis model as a whole. In this research, the segment number for the elastic beam is 50. The detailed research basis is depicted in our prior work [18]. The configuration of the rigid part at Point D could be described as:

$$\mathbf{g}_{st}(\boldsymbol{\theta}) = \exp\left(\hat{\zeta}_A\theta_A\right)\exp\left(\hat{\zeta}_B\theta_B\right)\exp\left(\hat{\zeta}_1\theta_1\right)\cdots\exp\left(\hat{\zeta}_n\theta_n\right)\mathbf{g}_{st,0} = \mathbf{g}_t \tag{1}$$

where $\mathbf{g}_{st,0}$ is the initial configuration of the tool frame {T} relative to the base frame {S}; $\mathbf{g}_t$ is the target configuration of rigid part at Point D after rotating each joint; θ_A and θ_B are the rotating angles of Point A and Point B relative to their initial angle; θ_i is the displacement of each elastic joint; $\hat{\zeta}_i$ is the joint twists for Points A and B, and points on the elastic beam at the initial configuration.

To solve the configuration problem for both position and angle at the tool frame, Equation (1) could be depicted below for further solution:

$$\left(\ln\left(\mathbf{g}_{st}\mathbf{g}_t^{-1}\right)\right)^\vee = 0 \tag{2}$$

The model also needs to meet the force balance constraint. The relation between the gripping force $\boldsymbol{F}_c$ and input torque for rigid-elastic hybrid mechanism could be expressed as:

$$M_d = -F_c * \sin(\alpha) * L_{de} \tag{3}$$

where M_d is the torque for rotating the rigid part at Point D; α is the angle of the parallelogram mechanism; L_{de} is the length of beam DE.

For the serial rigid-elastic hybrid mechanism, the force balance equation could be expressed as:

$$\boldsymbol{\tau} = \mathbf{K}_\theta\boldsymbol{\theta} - \mathbf{J}_D^T\mathbf{F}_D = \mathbf{0} \tag{4}$$

where $\mathbf{K}_{\boldsymbol{\theta}} = \mathrm{diag}(k_A, k_B, k_1, \cdots, k_n) \in \mathbb{R}^{(n+2)\times(n+2)}$ is the stiffness matrix of the joints; the stiffness of the elastic joints is set constant as k, while the stiffness of the normal joints like Joints A and B is set as zero; $F_D = \begin{bmatrix} f_{dx} \\ f_{dy} \\ M_d \end{bmatrix}$ is the force vector in plane, including force and torque; $\mathbf{J}_D \in \mathbb{R}^{3\times(n+2)}$ is the Jacobian matrix relating the joint space to the frame at Point D.

The geometric constraint and force balance equation for the rigid-elastic hybrid mechanism could be put into one matrix as:

$$\boldsymbol{C}(\boldsymbol{\theta}, \boldsymbol{F}_D) = \begin{bmatrix} \boldsymbol{y} \\ \boldsymbol{\tau} \end{bmatrix} = \begin{bmatrix} \left(\ln\left(\mathbf{g}_{st}\mathbf{g}_t^{-1}\right)\right)^{\vee} \\ \mathbf{K}_{\boldsymbol{\theta}}\boldsymbol{\theta} - \mathbf{J}_D^T \boldsymbol{F}_D \end{bmatrix} = \mathbf{0} \tag{5}$$

By differentiating the matrix, all the block elements could be expressed as a concise way. For the differentiating details of the components, we can refer to our prior work.

$$\nabla_C = \begin{bmatrix} \dfrac{\partial \boldsymbol{y}}{\partial \boldsymbol{\theta}} & \dfrac{\partial \boldsymbol{y}}{\partial \mathbf{F_D}} \\ \dfrac{\partial \boldsymbol{\tau}}{\partial \boldsymbol{\theta}} & \dfrac{\partial \boldsymbol{\tau}}{\partial \mathbf{F_D}} \end{bmatrix} = \begin{bmatrix} \mathbf{J}_D & \mathbf{0} \\ \mathbf{K}_{\theta} - \mathbf{K}_{\mathbf{J}_D} & -\mathbf{J}_D^T \end{bmatrix} \tag{6}$$

Based on Equations (6) and (5), we can use the Newton-Raphson method to obtain:

$$\begin{bmatrix} \boldsymbol{\theta} \\ \boldsymbol{F}_D \end{bmatrix}_{k+1} = \begin{bmatrix} \boldsymbol{\theta} \\ \boldsymbol{F}_D \end{bmatrix}_k + \left(\nabla_C\right)_k^{-1} \boldsymbol{C}_k \tag{7}$$

Thus, the elastostatic modeling of the gripper is completed.

The stiffness calibration process should be performed to get the value of k for the elastic joints mentioned before. Firstly, the value of k is substituted into the model, and the measured actuation amount and tip position are also put into the model. Then, we compare the calculated gripping force and the measured one to get the error. Finally, after accumulating the absolute value of errors in different sets of measured data and picking the value of k to make the minimum accumulated error, the k value is seen as the real value of joints in the elastic beam.

4. Model-Based Gripping Force Sensing and Strain Gauge Calibration

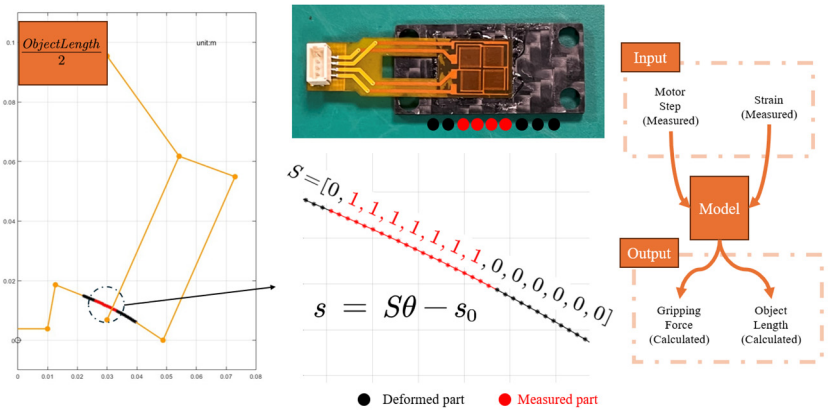


Figure 3. Methodology of gripping force sensing.

The model-based gripping force sensing uses the model discussed before and the strain obtained by the strain gauge to calculate the tip position and the gripping force, which is illustrated in Figure 3. The input of the model is actuation amount and deformation, and the output is gripping force and object

length. The bending angle of a deflection sensor linearly corresponds to the sum of elastic joints' displacements it covers. It can be expressed in a mathematical form as below:

$$\boldsymbol{s} = \boldsymbol{S}_e\boldsymbol{\theta} - \boldsymbol{s}_0 = \mathbf{0} \tag{8}$$

where $\boldsymbol{s}_0$ is the readings of the sensors; $\mathbf{S}_e$ depicts the position of the sensors, whose value is $\mathbf{S}_e = [0, \cdots, 0, 1, \cdots, 1, 0, \cdots, 0]$; $\boldsymbol{\theta}$ is the vector of the displacement angle for points on the elastic beam; $\boldsymbol{s}_0$ is the measured deformation of the region that the strain gauge covers. In the sensing model, the torque at Point D is seen as an unknown variable. The deformation is acquired by the strain gauge. So, the number of unknown variables is still equal to the number of constraint equations. The sensed information could be seen as a constraint to the model, and Equation (5) is transferred as:

$$C\left(\boldsymbol{\theta}, \boldsymbol{F}_D\right) = \begin{bmatrix} \boldsymbol{y} \\ \boldsymbol{\tau} \\ \boldsymbol{s} \end{bmatrix} = 0 \tag{9}$$

By differentiating the equation, the matrix is found as:

$$\nabla_C = \begin{bmatrix} \mathbf{J}_D & \mathbf{0} & \mathbf{0} \\ \mathbf{K}_\theta - \mathbf{K}_{\mathbf{J}_D} & -\mathbf{J}_D^T & \mathbf{0} \\ [0, 0, \mathbf{S}_e] & \mathbf{0} & \mathbf{0} \end{bmatrix} \tag{10}$$

Equation (9) could be solved by the Newton-Raphson method. Thus, given the actuation amount and the deformation, this equation could be solved to obtain the gripping force and gripping length.

For the calibration of a strain gauge, object length and actuation amount need to be measured and the theoretical deformation of the strain gauge is calculated. Then comparing the data of measured strain gauge value and the reading of strain gauge, the relation between the reading and the real deformation could be obtained.

5. Experimental Validation

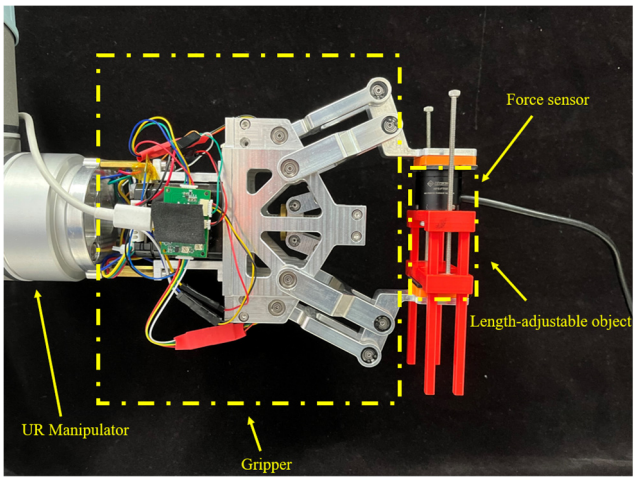


Figure 4. Experimental setup for validation.

The experiment setup is illustrated in Figure 4. The gripper was mounted on a UR robot to simulate the working environment. A length-adjustable object was used to provide different lengths and tip positions for validation. The force sensor was mounted on the length-adjustable object to record the gripping force during gripping. The object length, motor actuation amount, gripping force, and strain of the deformable part could all be measured with this set of experiment equipment. Then, the theory could be validated by experiments.

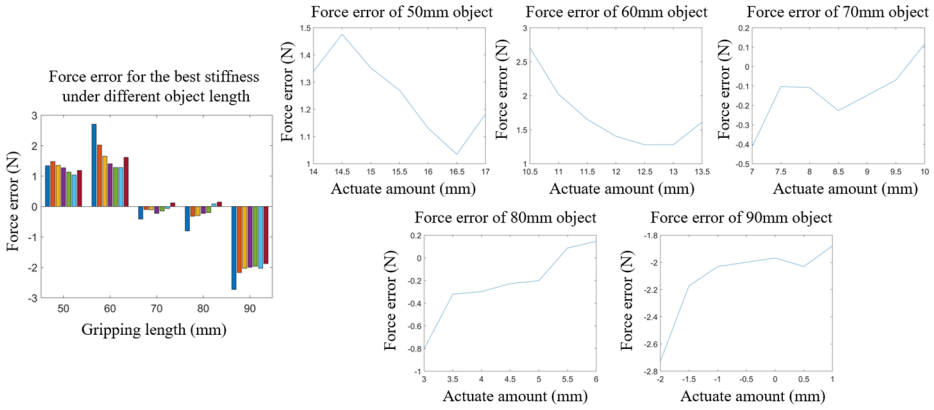


Figure 5. Validation error of stiffness.

By implementing different actuate amounts on the same length of the object, the error between the calculated force and measured force could be calculated. Then, the overall performance could be gained by changing the length of objects and taking records. Taking the RSME of the error, the best stiffness could be determined and Figure 5 shows the error of calculation force for gripping objects with different gripping object lengths for the best stiffness.

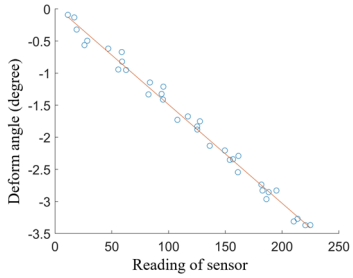


Figure 6. Strain calibration.

The relation between the deform angle and the reading of the sensor could be determined by the strain calibration mentioned before. The result is shown in Figure 6. Using the fitting method for simple linear regression, the relation could be found as a red line in Figure 6.

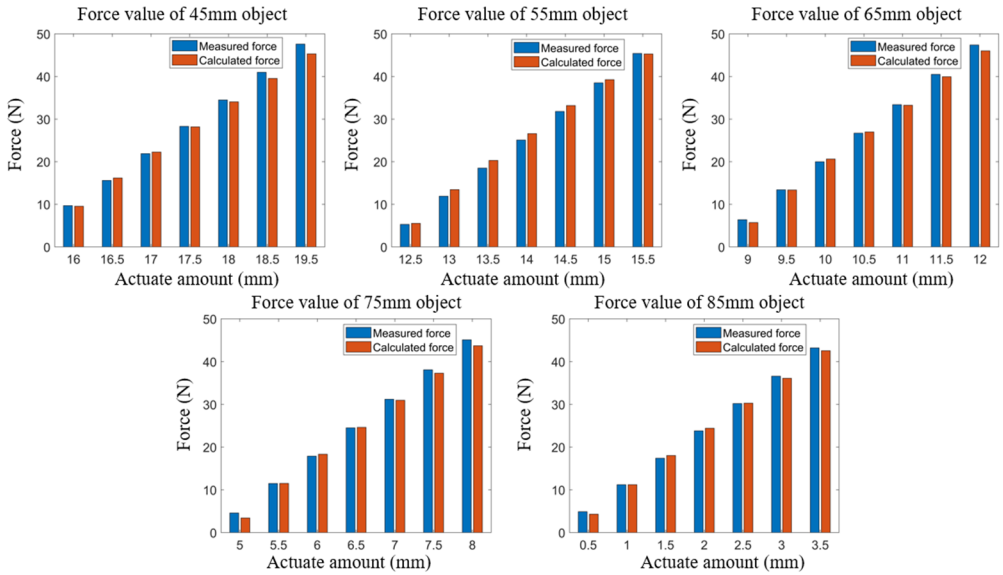


Figure 7. The force value of the calculated force and measured force.

We taking object length that is different from the calibration process for force sensing validation. The force calculation error is depicted in Figure 7. The root mean square error (RSME) of the force sensing experiments is 0.893.

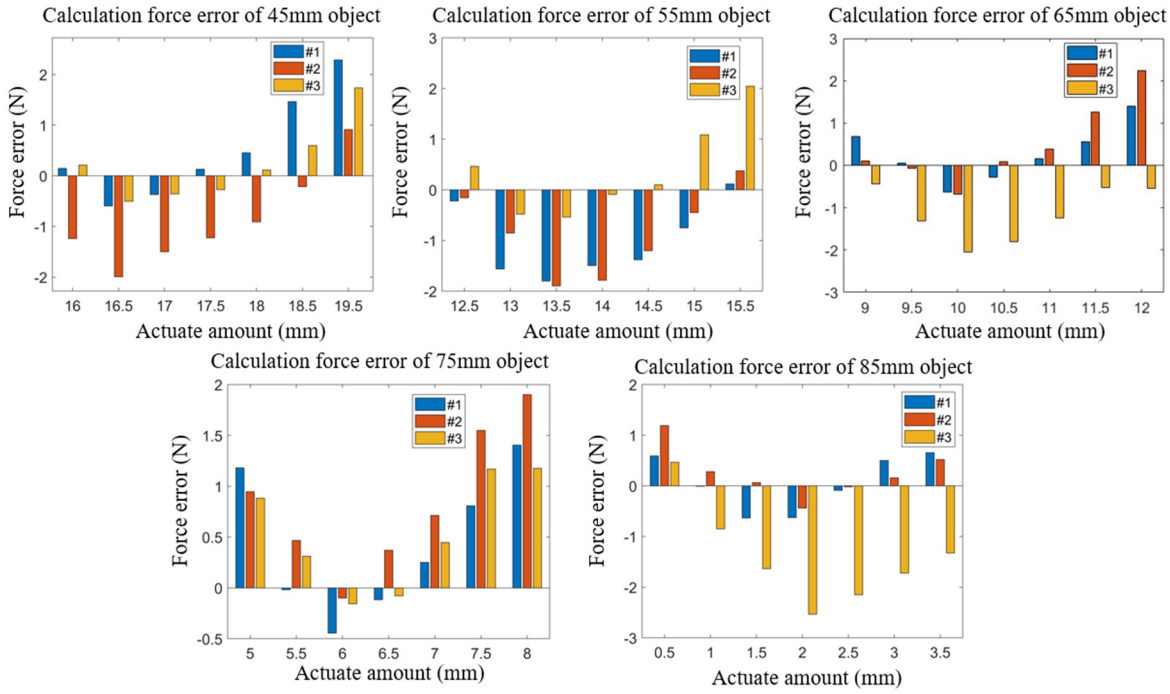


Figure 8. Force error in repeated experiments.

To validate the repeatability of the force calculation method, repeated experiments were conducted. The force error was collected three times for different gripping lengths under the same circumstances, and the results were recorded in Figure 8. The result shows that the maximum force error is 2.5 N and the Mean Absolute Error (MAE) is 0.7957 N.

6. Conclusion

Based on the results and discussions presented above, the conclusions are obtained as below:

(1) It is shown that the mean error of the force sensing is 2.5 N, with the maximum gripping force of 47.6 N.

(2) The Mean Absolute Error in the repeating test is 0.7957 N. This shows the high accuracy of the proposed model and its stability.

(3) The proposed force-sensing gripper could be put into use in industrial automation to realize the gripping force control in gripping large pieces of the workpiece in the automation of manufacturing.

Acknowledgments

This work was financially supported by the National Key Research and Development Program of China under Grant 2019YFA0709001.

References

[1] Z. Zhu, Y. Liu, J. Ju, and E. Lu, “Design and Experimental Test of Rope-Driven Force Sensing Flexible Gripper”, Sensors, Vol. 24, No. 19, p. 6, 407, Oct. 2024, doi: 10.3390/s24196407.
[2] Y. Yan, S. Guo, C. Lyu, D. Zhao, and Z. Lin, “SEA-Based Humanoid Finger-Functional Parallel Gripper with Two Actuators: PG2 Gripper”, IEEE Trans. Instrum. Meas., Vol. 72, pp. 1-13, 2023, doi: 10.1109/TIM.2022.3229695.

[3] J. Zhou, Y. Chen, X. Chen, Z. Wang, Y. Li, and Y. Liu, “A Proprioceptive Bellows (PB) Actuator with Position Feedback and Force Estimation”, IEEE Robot. Autom. Lett., Vol. 5, No. 2, pp. 1, 867-1, 874, Apr. 2020, doi: 10.1109/LRA.2020.2969920.
[4] M. S. Li, T. M. Huh, C. R. Yahnker, and H. S. Stuart, “Resonant Pneumatic Tactile Sensing for Soft Grippers”, IEEE Robot. Autom. Lett., Vol. 7, No. 4, pp. 10, 105-10, 111, Oct. 2022, doi: 10.1109/LRA.2022.3191186.
[5] Y. C. Pei, B. H. Wang, J. T. Wu, C. Wang, J. H. Guan, and H. Lu, “A Machine Learning Empowered Shape Memory Alloy Gripper with Displacement-Force-Stiffness Self-Sensing”, IEEE Trans. Ind. Electron., Vol. 70, No. 10, pp. 10, 385-10, 395, Oct. 2023, doi: 10.1109/TIE.2022.3222655.
[6] W. Xu, H. Zhang, H. Yuan, and B. Liang, “A Compliant Adaptive Gripper and Its Intrinsic Force Sensing Method”, IEEE Trans. Robot., Vol. 37, No. 5, pp. 1, 584-1, 603, Oct. 2021, doi: 10.1109/TRO.2021.3060971.
[7] Y. Du, P. Zhang, and T. Li, “Contact force measurement of the compliant gripper based on a force sensing model”, Rev. Sci. Instrum., Vol. 95, No. 12, p. 125, 107, Dec. 2024, doi: 10.1063/5.0238161.
[8] J. Peng, W. Li, L. Liu, W. Liu, and Y. Han, “Design, Perceptual Modeling, and Grasping Performance Evaluation of Multibranch Flexible Grippers”, IEEE Trans. Instrum. Meas., Vol. 74, pp. 1-18, 2025, doi: 10.1109/TIM.2025.3529055.
[9] A. N. Reddy, N. Maheshwari, D. K. Sahu, and G. K. Ananthasuresh, “Miniature Compliant Grippers with Vision-Based Force Sensing”, IEEE Trans. Robot., Vol. 26, No. 5, pp. 867-877, Oct. 2010, doi: 10.1109/TRO.2010.2056210.
[10] J. Yi, B. Kim, K. J. Cho, and Y. L. Park, “Underactuated Robotic Gripper with Fiber-Optic Force Sensing Tendons”, IEEE Robot. Autom. Lett., Vol. 8, No. 11, pp. 7, 607-7, 614, Nov. 2023, doi: 10.1109/LRA.2023.3315204.
[11] J. Huang, Z. Wang, J. Li, and Z. Tang, “A design of underwater soft gripper with water pressure sensing and enhanced stiffness”, Robotica, Vol. 41, No. 2, pp. 548-565, Feb. 2023, doi: 10.1017/S0263574722001412.
[12] C. Tang, X. Chang, J. Wang, Y. Peng, H. Wu, and H. Wang, “A Nonarray Soft Capacitive Tactile Sensor with Simultaneous Contact Force and Location Measurement for Intelligent Robotic Grippers”, IEEE Trans. Instrum. Meas., Vol. 73, pp. 1-9, 2024, doi: 10.1109/TIM.2023.3343795.
[13] J. Hashizume, T. M. Huh, S. A. Suresh, and M. R. Cutkosky, “Capacitive Sensing for a Gripper with Gecko-Inspired Adhesive Film”, IEEE Robot. Autom. Lett., Vol. 4, No. 2, pp. 677-683, Apr. 2019, doi: 10.1109/LRA.2019.2893154.
[14] J. Zhang, S. Lai, H. Yu, E. Wang, X. Wang, and Z. Zhu, “Fruit Classification Utilizing a Robotic Gripper with Integrated Sensors and Adaptive Grasping”, Math. Probl. Eng., Vol. 2021, pp. 1-15, Sep. 2021, doi: 10.1155/2021/7157763.
[15] Q. Xu, “Design and Development of a Novel Compliant Gripper with Integrated Position and Grasping/Interaction Force Sensing”, IEEE Trans. Autom. Sci. Eng., Vol. 14, No. 3, pp. 1, 415-1, 428, Jul. 2017, doi: 10.1109/TASE.2015.2469108.
[16] T. Guan, S. Tang, G. Chen, W. Yan, J. Luo, and H. Wang, “Design and Development of a Composite Compliant Two-finger Gripper”, in Intelligent Robotics and Applications, Vol. 14, 268, H. Yang, H. Liu, J. Zou, Z. Yin, L. Liu, G. Yang, X. Ouyang, and Z. Wang, Eds., in Lecture Notes in Computer Science, Vol. 14, 268, Singapore: Springer Nature Singapore, 2023, pp. 558-567. doi: 10.1007/978-981-99-6486-4_47.
[17] G. Chen et al., “Intrinsic Contact Sensing and Object Perception of an Adaptive Fin-Ray Gripper Integrating Compact Deflection Sensors”, IEEE Trans. Robot., Vol. 39, No. 6, pp. 4, 482-4, 499, Dec. 2023, doi: 10.1109/TRO.2023.3311610.
[18] G. Chen, Z. Zhang, and H. Wang, “A General Approach to the Large Deflection Problems of Spatial Flexible Rods Using Principal Axes Decomposition of Compliance Matrices”, J. Mech. Robot., Vol. 10, No. 3, p. 031012, Jun. 2018, doi: 10.1115/1.4039223.